Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

8DDAPM Module: -

The RS485 IO module is a compact, robust, and low-power device designed for reliable communication over distances up to 700 meters without a repeater, depending on the signal strength of the Modbus master. It features 8 digital inputs, 8 digital outputs, 8 analog inputs (10-bit resolution, supporting 0–20 mA and 0–10 V), and 8 RTD (PT1000) channels, making it ideal for versatile industrial applications. Housed in a DIN rail-mountable enclosure, the module includes exposed connectors for easy wiring and LED indicators for real-time I/O status monitoring. Configuration of slave ID and baud rate is done via internal DIP switches. To enhance system protection, the module is equipped with resettable fuses guarding against reverse polarity on both power and RS485 communicatio

Specifications

General –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	155 mm L x 110 mm B x 75 mm H
Power	Input Power – 12 - 24 VDC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Certifications	RoHS, CE APPROVED

DI Inputs –

Channels	8
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Digital Output

Channels	8
Output Type	P-Channel MOSFET
Channel Voltage	12 VDC/24 VDC
Channel Current	20A
Channel Power	240 W (max)
Operating Time	2.8ns.
Turn Off delay Time	33.5ns.
Operating Temperature	-55℃~+150

AI Inputs –

Channels	8
Input Signal	4 – 20 mA / 0 – 10VDC
Accuracy	± 2% Full scale
Input Resolution	10 bits
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

AI Inputs –

Channels	8– PT1000 – (jumper selectable) three wire
Accuracy	Total Accuracy over All Operating Conditions: 0.5NC (0.05% of Full Scale) max.
Input Resolution	15-Bit ADC Resolution.
Isolation	Optically Isolated.

Additional Features: -

All inputs and communication ports isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
No configuration needed on the IO board

Configuration Settings: -

Communication Speed	9600 – 115200 (DIP SW selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Configurable with DIP Switch
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3,4,5,6,7.
Function code DI	0x02 Read discrete Input
DI Register Address	8, 9,10,11,12,13,14,15.
Function Code AI	0x03 Read Holding Registers
AI Register Address	10 Bit – 1, 2, 3, 4, 5, 6, 7, 8.
RTD Register Address	9, 10, 11, 12, 13,14,15,16.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	**DO 1**	00001	0X05,0X0F	RS485	1 Bit Boolean
2	**DO 2**	00002	0X05,0X0F	RS485	1 Bit Boolean
3	**DO 3**	00003	0X05,0X0F	RS485	1 Bit Boolean
4	**DO 4**	00004	0X05,0X0F	RS485	1 Bit Boolean
5	**DO 5**	00005	0X05,0X0F	RS485	1 Bit Boolean
6	**DO 6**	00006	0X05,0X0F	RS485	1 Bit Boolean
7	**DO 7**	00007	0X05,0X0F	RS485	1 Bit Boolean
8	**DO 8**	00008	0X05,0X0F	RS485	1 Bit Boolean
9	**DI 1(With & Without Potential)**	10009	0X02	RS485	1 Bit Boolean
10	**DI 2(With & Without Potential)**	10010	0X02	RS485	1 Bit Boolean
11	**DI 3(With & Without Potential)**	10011	0X02	RS485	1 Bit Boolean
12	**DI 4(With & Without Potential)**	10012	0X02	RS485	1 Bit Boolean
13	**DI 5(With & Without Potential)**	10013	0X02	RS485	1 Bit Boolean
14	**DI 6(With & Without Potential)**	10014	0X02	RS485	1 Bit Boolean
15	**DI 7(With & Without Potential)**	10015	0X02	RS485	1 Bit Boolean
16	**DI 8(With & Without Potential)**	10016	0X02	RS485	1 Bit Boolean
17	**AI 1**	40002	0X03	RS485	16 Bit Unsigned int
18	**AI 2**	40003	0X03	RS485	16 Bit Unsigned int
19	**AI 3**	40004	0X03	RS485	16 Bit Unsigned int
20	**AI 4**	40005	0X03	RS485	16 Bit Unsigned int
21	**AI 5**	40006	0X03	RS485	16 Bit Unsigned int
22	**AI 6**	40007	0X03	RS485	16 Bit Unsigned int
23	**AI 7**	40008	0X03	RS485	16 Bit Unsigned int
24	**AI 8**	40009	0X03	RS485	16 Bit Unsigned int
25	**RTD1 1000**	40010	0X03	RS485	16 Bit Unsigned int
26	**RTD2 1000**	40011	0X03	RS485	16 Bit Unsigned int
27	**RTD3 1000**	40012	0X03	RS485	16 Bit Unsigned int
28	**RTD4 1000**	40013	0X03	RS485	16 Bit Unsigned int
29	**RTD5 1000**	40014	0X03	RS485	16 Bit Unsigned int
30	**RTD6 1000**	40015	0X03	RS485	16 Bit Unsigned int
31	**RTD7 1000**	40016	0X03	RS485	16 Bit Unsigned int
32	**RTD8 1000**	40017	0X03	RS485	16 Bit Unsigned int

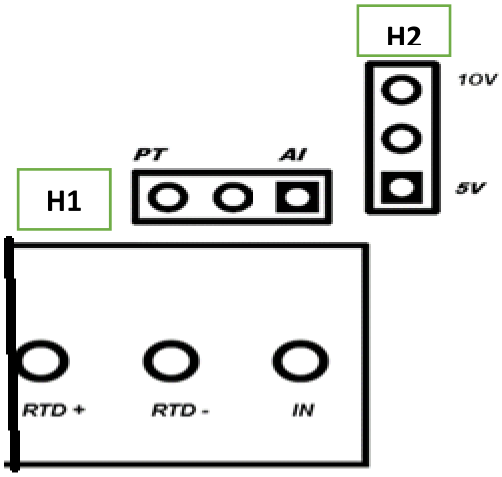
Note: -

For MODBUS communications, **a shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

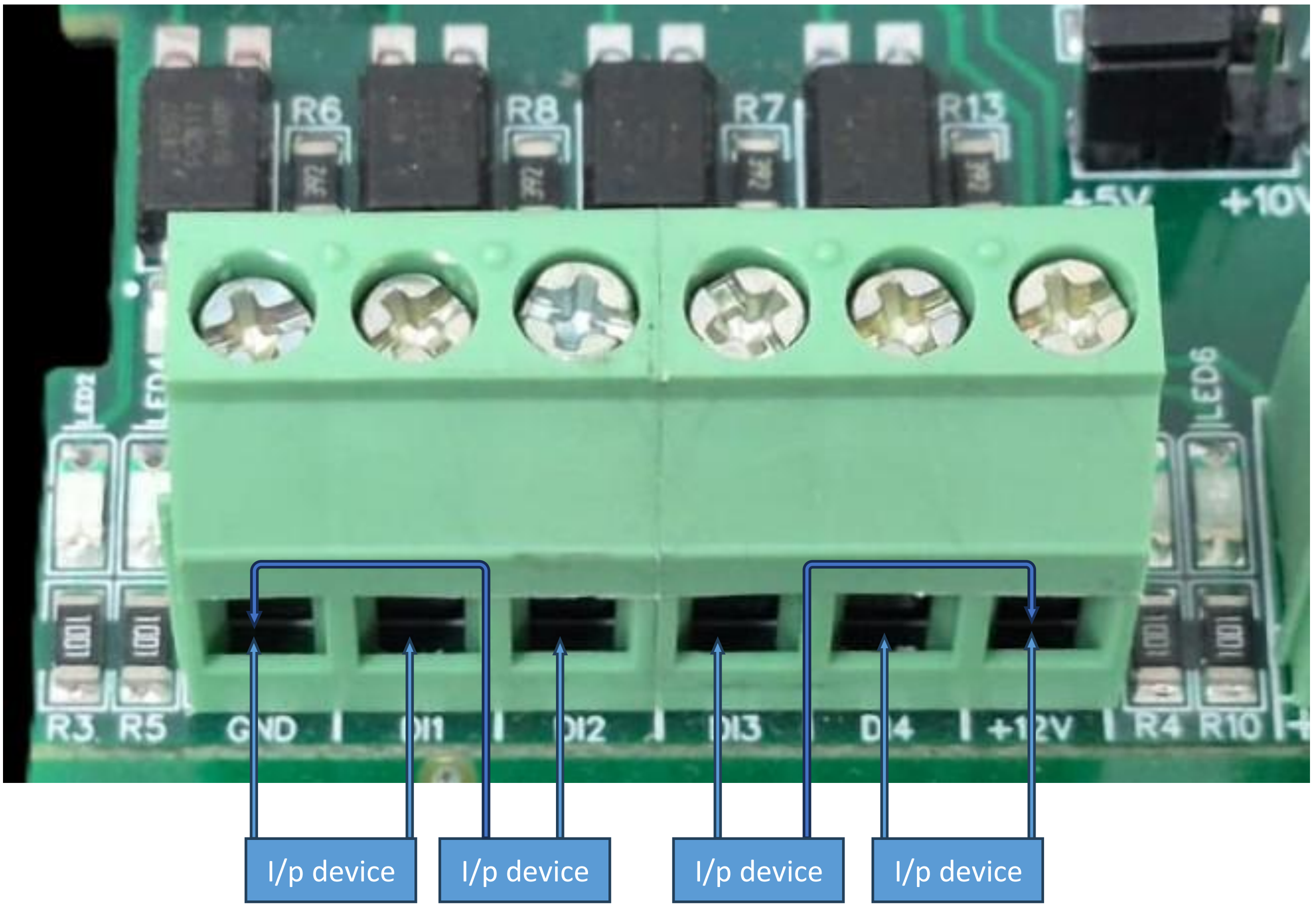
Connection Instruction FOR WIN - IO – 8DDAPM



Note:

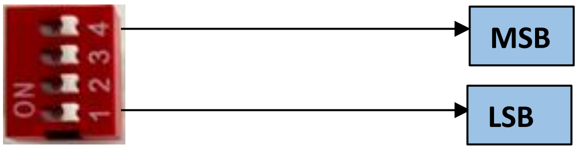
- Identify the AI/PT1000 wiring, configuration, or specification
- For AI there are 2 Jumpers Eg. Jumper “H1” & “H2” are for Selecting the Input Voltage Detection. + & - Terminal are used for all AI INPUT
  The system supports monitoring of both Analog Input (AI) and RTD (PT1000) sensor data.
- Selection between AI and RTD input modes is achieved by configuring the appropriate jumper settings. Refer to the provided diagram for correct jumper positions and wiring connections.
- Ensure that only one mode is selected at a time to prevent signal conflict or inaccurate readings.
- H1 is for selection of analog input and RTD(PT1000)
- H2 is for voltage selection between 5V and 10V.

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
  - **GND:** Common Ground reference.
  - **DI1 – DI4:** Individual Digital Input channels.
  - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
  - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
  - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

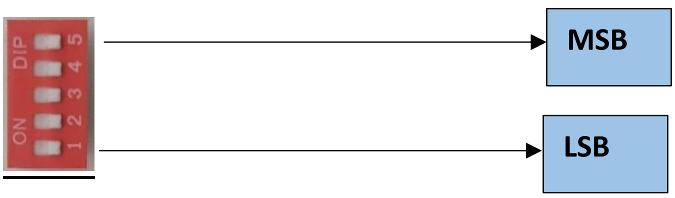
BAUD RATE DESCRIPTION



- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When u change the Baud rate in the Module power 'ON' condition, pls press the reset button to get Change to affect.

Baud Rate	DIP SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
115200	ON	OFF	OFF	OFF
57600	OFF	ON	OFF	OFF
38400	OFF	OFF	ON	OFF
19200	OFF	OFF	OFF	ON

SLAVE ID DESCRIPTION



For Slave ID Selection SW is used to Set The SLAVE ID .
For Slave ID DIP Switch LSB is "1" follow through "5" is MSB.
Slave ID Confirmed through below Device ID table .
IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So"0 0 0 0 1" will be selected as Slave ID 1.

Slave ID Configuration

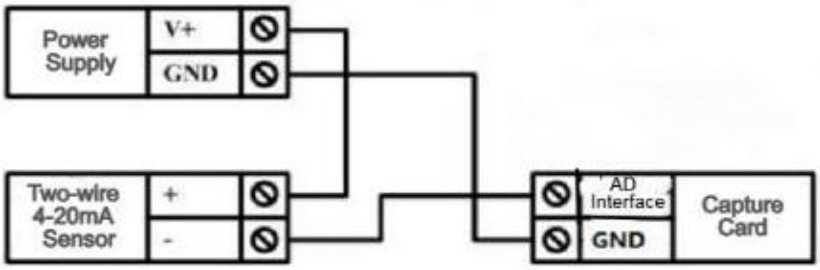
The slave ID can be configured as per user preference using a 5-bit DIP switch setting. The binary value set by the switches corresponds to a specific decimal slave ID.

Binary (DIP Switch)	Decimal Slave ID
00001	1
10000	16
11111	31

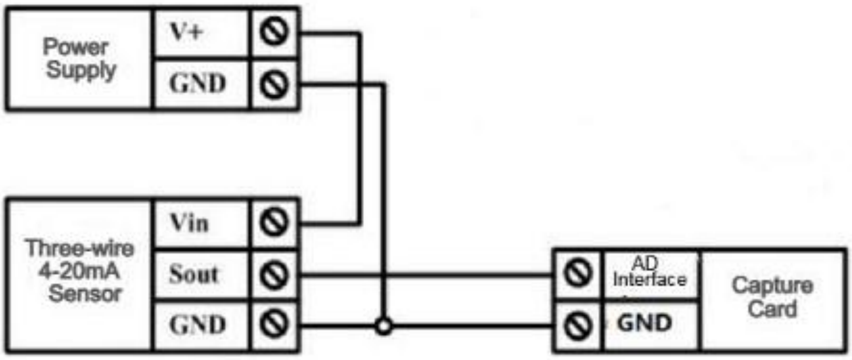
Note:

- Bit position and value mapping may vary based on hardware implementation (e.g., LSB ↔ MSB). Refer to the device's DIP switch bit order specification when setting the ID.
- DIP switches **SW1 to SW5** are provided for user configuration. These switches can be set as per the application requirements to control functions such as **slave address selection**, **mode settings**, or other device parameters.
- The device supports configurable slave addresses within the range of **1 to 31**, which can be set according to application requirements.

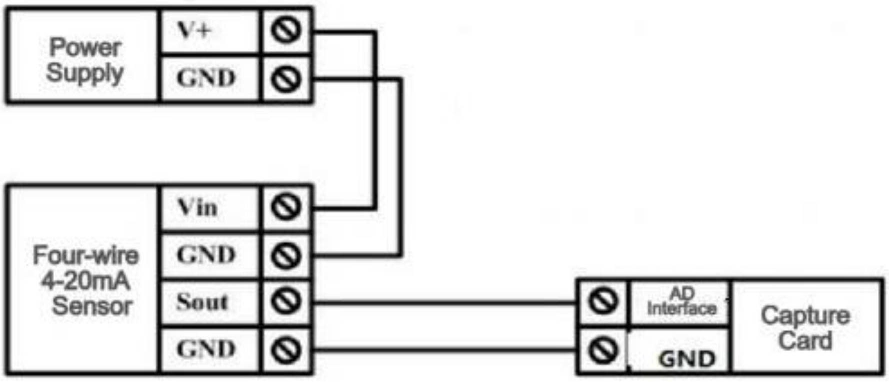
Two-wire Sensor Wiring Diagram

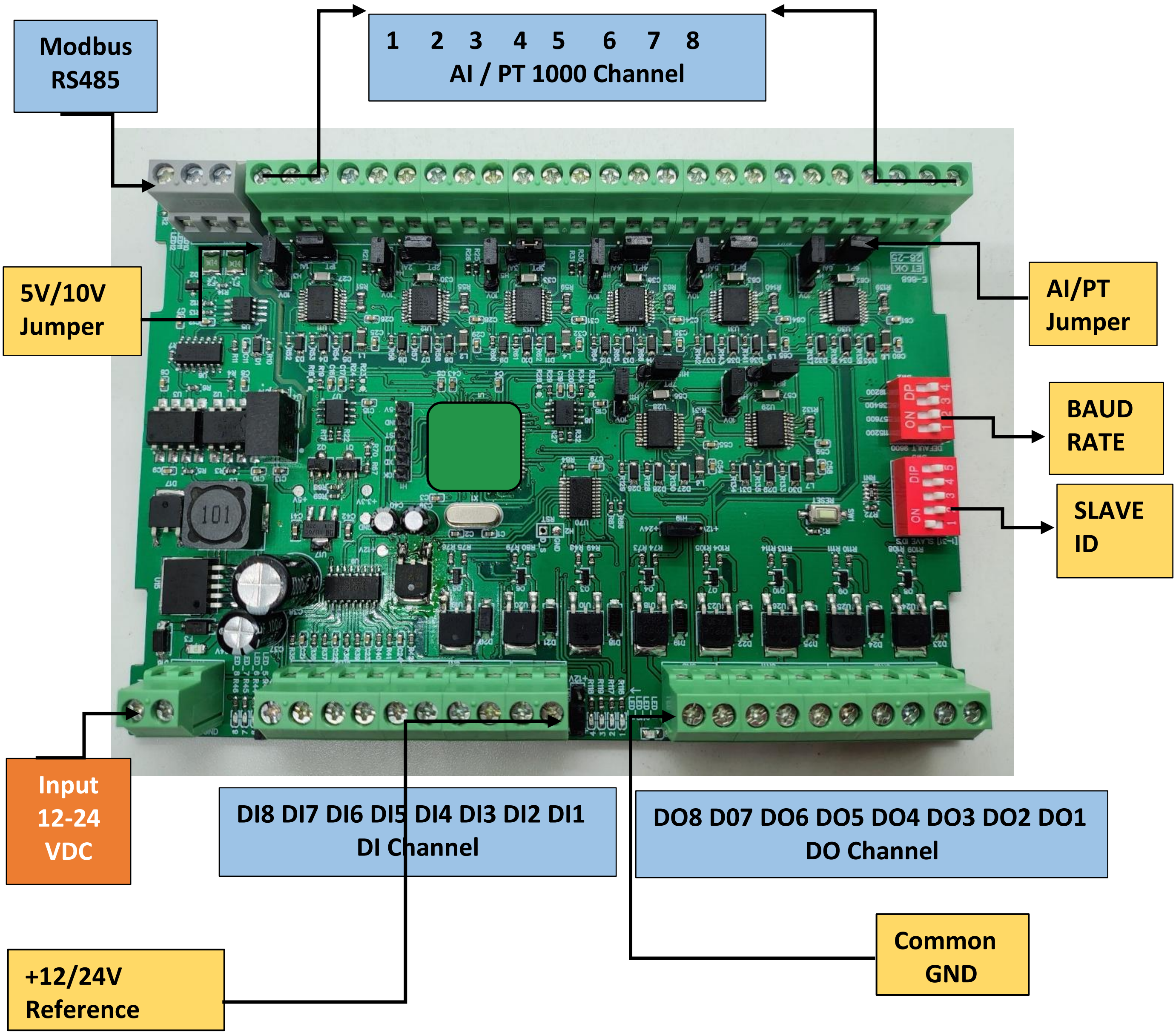


Three-wire Sensor Wiring Diagram



Four-wire 4-20mA Sensor





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